

FITBUDDY - AI FITNESS PLAN GENERATOR USING GEMINI MODELS

Phase 6: Project Testing Phase

This phase verifies that all components work correctly and that the application provides the expected results.

Functional Testing

The following features are tested:

- User registration/input.
- Workout plan generation.
- Nutrition tip generation.
- Feedback submission.
- Workout plan updating.
- Database storage.
- User retrieval.
- Administrative user view.

API Testing

FastAPI's interactive documentation can be used to test endpoints through:

`http://127.0.0.1:8000/docs`

UI Testing

The following pages are tested:

- Home page.
- Personalized workout page.
- Nutrition tip section.
- Feedback page.
- Updated plan display.
- View All Users page.

Database Testing

Testing verifies that:

- User details are stored correctly.
- Original plans are preserved.
- Updated plans are stored correctly.
- User and plan information can be retrieved.

Error Testing

The application is checked for:

- Invalid form input.
- Missing user information.
- Invalid User ID.
- API failures.
- Database errors.
- Empty feedback submissions.

Responsive Testing

The frontend is tested on different screen sizes to verify mobile and desktop compatibility.

Outcome:

A tested and verified application with identified and resolved functional or technical issues.